

Competency frameworks and standards for digital health

A landscape analysis



World Health
Organization

Competency frameworks and standards for digital health

A landscape analysis



**World Health
Organization**

Competency frameworks and standards for digital health: a landscape analysis

ISBN 978-92-4-011976-5 (electronic version)

ISBN 978-92-4-011977-2 (print version)

© World Health Organization 2026

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; <https://creativecommons.org/licenses/by-nc-sa/3.0/igo>).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: "This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition".

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (<http://www.wipo.int/amc/en/mediation/rules/>).

Suggested citation. Competency frameworks and standards for digital health: a landscape analysis. Geneva: World Health Organization; 2026. Licence: [CC BY-NC-SA 3.0 IGO](https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

Cataloguing-in-Publication (CIP) data. CIP data are available at <https://iris.who.int/>.

Sales, rights and licensing. To purchase WHO publications, see <https://www.who.int/publications/book-orders>. To submit requests for commercial use and queries on rights and licensing, see <https://www.who.int/copyright>.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Contents

Acknowledgements	iv
Abbreviations and acronyms	v
Executive summary	vi
1. Introduction	1
2. Methods	3
3. Findings	5
3.1 Definition of a digital health competency framework	6
3.2 Characteristics of digital health competency frameworks	6
3.3 Mapping competencies across frameworks	7
3.4 Methods employed for framework development in the included studies	9
4. Discussion	11
4.1 Fostering acquisition of digital health competencies	12
4.2 Limitations	15
5. Conclusion	17
References	19
Annex 1: Study characteristics	23
Annex 2: Overview of competency frameworks	29
Annex 3: List of studies and corresponding competency domains/ categories/areas	33
Annex 4: Framework mapping of corresponding domains/ categories/areas	37

Acknowledgements

The development of this technical report was undertaken by the World Health Organization (WHO) Collaborating Centre for Digital Health and Health Education, at the Centre for Population Health Sciences at the Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore, under the leadership and guidance of James Campbell (Director, Health Workforce Department, WHO). Technical oversight and quality assurance were provided by Giorgio Cometto (Health Workforce Department, WHO) and Siobhan Fitzpatrick (Health Workforce Department, WHO). Project coordination and written contributions were provided by Onyema Ajuebor (Health Workforce Department, WHO).

WHO is grateful to the primary writers and contributors at the WHO Collaborating Centre for Digital Health and Health Education, under the leadership and guidance of Professor Josip Car. Other members of the team include Chin-Siang Ang, Lorainne Tudor Car, Sakura Ito, Jintana Liu, Dewan Mohammad Sumsuzzman and Tatiana Erlikh Fox.

WHO is also grateful to the following individuals for contributing their time and expertise to review the document at different stages of its development: Vajira HW Dissanayake (University of Colombo, Sri Lanka); Seble Frehywot (George Washington University, United States of America); Anna Frellsen (Maternity Foundation, Denmark); Taghreed Issam Justinia (King Saud bin Abdulaziz University for Health Sciences, Saudi Arabia); Wendy Leonard (TIP Global Health, Rwanda); Andrew Likaka (Independent consultant, Malawi); Jodi Lis (IntraHealth International, United States of America); Ahmed Mandil (University of Alexandria, Egypt); Alvin B Marcelo (University of the Philippines, Manila); Semee Hwangbo Pak (National Academy of Sciences, United States of America); Jai Ganesh Udayasankaran (Asia eHealth Information Network, India).

The potential of conflict of interest was assessed and managed according to WHO policy.

WHO is grateful to colleagues who provided insightful comments during the development and review of this report; they include: Pascoal Bento (Data, Digital Health, Analytics and AI Department, WHO); Clayton Hamilton (Data and Digital Health, WHO Regional Office for Europe); Khassoum Diallo (Health Workforce Department, WHO); Catherine Kane (Health Workforce Department, WHO); Michelle McIsaac (Health Workforce Department, WHO); Derrick Muneene (Data, Digital Health, Analytics and AI Department, WHO); Diana Zandi (Performance, Financing and Delivery Department, WHO); Pascal Zurn (Health Workforce Department, WHO).

Financial support for the development and dissemination of this guidance was provided by the European Commission, the Spanish Agency for International Cooperation and Development (AECID) and the South Korea Ministry of Health and Welfare.



Abbreviations and acronyms

AI	artificial intelligence
EHR	electronic health records
HETA	health workforce education technology assessment
ICT	information and communication technologies
NT-EPAs	nursing telehealth entrustable professional activities
SDG	Sustainable Development Goal
TIGER	Technology Informatics Guiding Education Reform (framework)
WHO	World Health Organization



Executive summary

The increasing adoption of digital technologies in health workforce education and practice underscores the need to strengthen health workers' digital competence. Recent public health challenges, notably the COVID-19 pandemic, displacements due to adverse weather conditions and growing conflict situations among others, have raised the demand for telemedicine and other remote health and learning services. This in turn has further fuelled the need for more efficient and scalable digital tools. These tools have become crucial as countries bolster their health and educational systems to accelerate the realization of universal health coverage, on the path towards achieving the health-related Sustainable Development Goals (SDGs) by 2030. To fully leverage the potential and benefits of digital tools, users must possess the necessary skills to use them effectively and the competencies to apply these skills in practice. Digital health competency frameworks serve as valuable reference tools that assist teaching faculty members and policy-makers responsible for developing learning programmes in obtaining a thorough understanding of the skills and competencies required for health workers' effective practice in health service delivery. However, digital health competency frameworks are often not consistently developed, adapted or used appropriately. Various factors contribute to this, including challenges in defining or specifying relevant digital health competencies within learning programmes. Consequently, it is imperative that leaders of health training institutions and policy-makers are cognisant of the latest evidence on digital health competency frameworks to help shape curricula and learning programmes that equip health workers with the requisite digital health competencies. Synthesizing evidence on the characteristics and use of digital health competency frameworks is an essential step in reinforcing health workforce education on digital health applications and their use in practice.

While some digital health competencies are applicable across various health professions, others may be more specific, depending on factors such as task complexity and the educational level of the learner. Moreover, the regulation of education and training programmes, as well as ensuring the consistency of the available technical infrastructure, are the main system-level requirements for fostering the acquisition of digital health competencies by the health workforce. Organizational support needs include clear identification of required digital health competencies, the development of efficient training programmes, and the provision of necessary digital learning and practice modalities tailored to local and institutional needs. This report maps out 10 competency clusters from the literature that lay the foundations for digital health competence. It also incorporates a set of policy options for consideration by policy-makers when developing or revising their frameworks, across three interrelated factors: change; adaptation; and sustainability. The overarching aim is to support policy- and decision-makers in their efforts to develop or adapt digital health competency frameworks within the educational and employment sectors, thereby enhancing the digital health competence of the health workforce.



1. Introduction

Digital health, formerly referred to as eHealth, is an umbrella term covering mobile health (mHealth) and other computing areas applicable to health, such as the Internet of Things and artificial intelligence (AI) (1). WHO defines digital health as “the use of information and communication technologies to support health and health-related fields”, including health service provision, monitoring, literature, education, knowledge and research. It also covers emerging areas such as AI, big data and genomics (2). Digital health technologies enable practitioners to track and monitor health using apps, mobile devices, sensors and wearables (2), contributing to improved quality, safety and efficiency of health services (3). This improvement is increasingly recognized in the literature and linked to broader developmental goals, such as those outlined in the health-related SDGs: SDG 3 – good health and well-being; SDG 4 – quality education; and SDG 9 – industry, innovation and infrastructure (4).

To harness digital health’s full potential, users must possess the necessary skills and competencies for the effective use of digital tools and technologies. This need is reflected by the WHO *Global strategy on digital health 2020–2025*, which provides policy options to strengthen health workers’ digital competencies (5). WHO has also identified the development of a digital health competency framework as a priority resource to be developed through the coordination of the Global Initiative on Digital Health (6). This is increasingly relevant as digitally competent health workers are much needed in the health labour market, with reports of up to 90% of health roles requiring digital skills in some settings (7). Concurrently, the importance of health digitization has grown, partly due to the surge in demand for remote care following the COVID-19 pandemic and an increase in human displacements due to adverse weather conditions and conflicts, among other reasons. Health workforce training institutions have also recognized the need to maintain continuous learning amidst such disruptions, leading to the swift adoption of digital tools for distance learning. It is also true that accessibility to digital tools remains inequitable, and this challenge needs to be addressed even as digital health competency framework resources are developed. As these technological learning and practice devices and tools have now become indispensable (8–10), improving health workers’ digital health competence is critical for ongoing digitization of health services and learning processes and materials (10–12).

This report is primarily intended for the heads and faculty members of health training institutions, as well as health policy-makers involved in developing or adapting competency frameworks or standards to deliver competency-based learning on digital health to health practitioners or to support practice using digital tools. United Nations agencies and other partners working to advance digital health competencies and skills may also add value to their work using its findings. The essence of this document lies in its examination and consolidation of the digital health competency framework space within health workforce education and practice. It adopts a scientific approach to describe how its findings on digital health competency frameworks can be better utilized to strengthen health workforce digital competence and applications in health service delivery. Furthermore, it outlines end-user enablers at the organizational and system levels and discusses how applying digital health competency frameworks at global and national levels can enable the achievement of digital health competence at the individual level. The report also proposes policy options that introduce how the concepts of change and sustainability can be embedded to support the application of digital health competency frameworks, noting the strategies and critical success factors involved.



2. Methods

The procedure carried out in this narrative review followed these steps:

1. Identification of relevant literature
2. Selection
3. Extraction
4. Compilation of thematic ideas.

A narrative review was chosen for this study to allow a broader scope of searching while maintaining methodological rigour. The search was performed across several databases, including MEDLINE, Embase, CINAHL (Cumulative Index to Nursing and Allied Health Literature), ERIC (Education Resources Information Center), PsycINFO and the Cochrane Library, spanning the years 2012 to 2022.

Search terms were adapted from a published review with a similar objective (12), intentionally kept general, including two main concepts: “digital health” and “competency framework”. Inclusion criteria specified that the digital health competency frameworks should target pre-service and in-service health workers in the fields of medicine, nursing and allied health, and be published in English. Articles not meeting these criteria were excluded.

Upon completing the search, all identified citations were imported into Endnote software, and duplicates were removed.

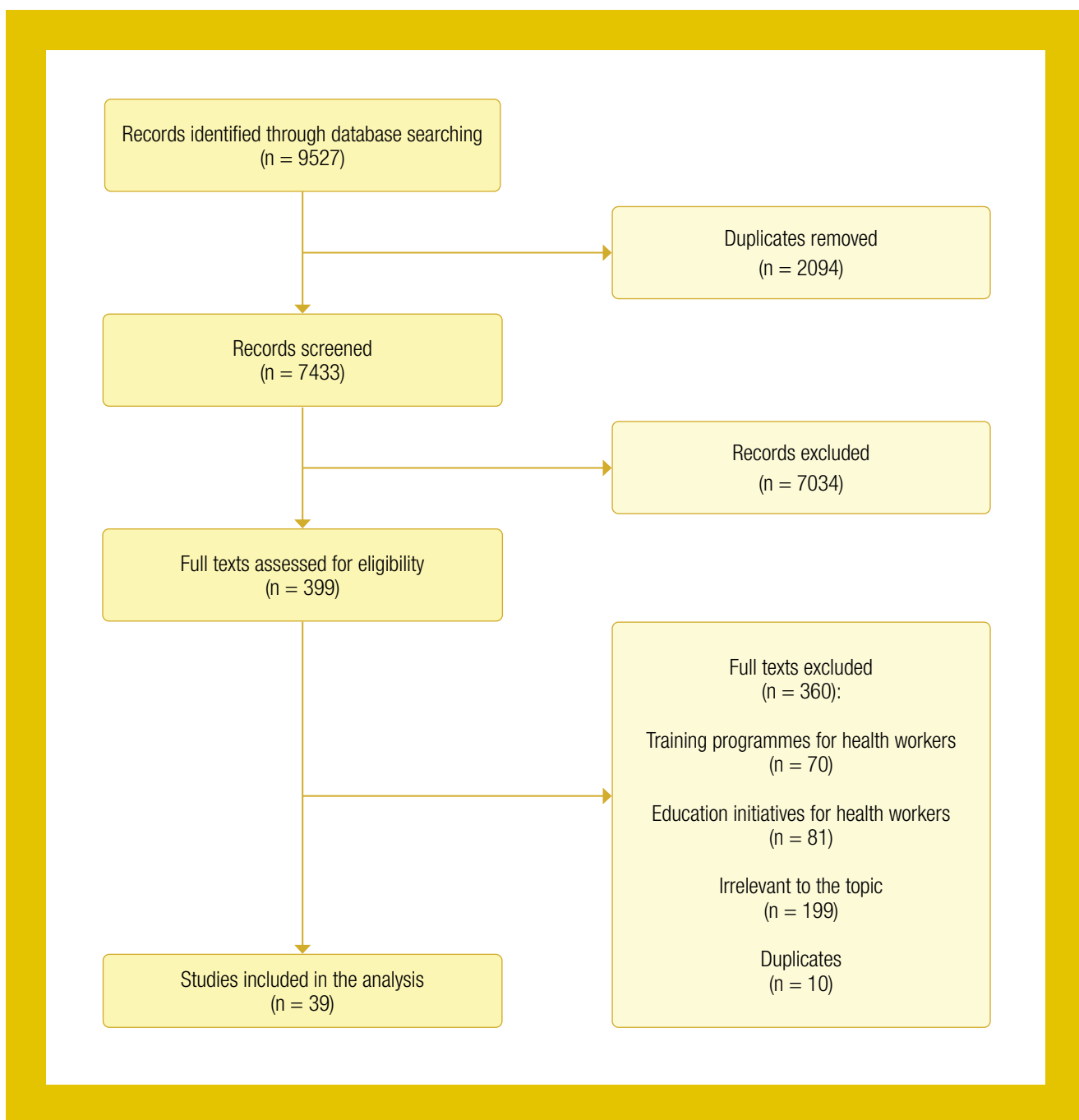
Titles and abstracts were independently screened by four reviewers, followed by full-text reviews. Disagreements between reviewers regarding study inclusion were resolved through discussion to reach a consensus.

Data extracted included study characteristics, framework characteristics and main findings. Findings were narratively synthesized and reported.

3. Findings

The search strategy yielded 9527 initial results, of which 2094 were identified as duplicates and subsequently removed. After a meticulous screening process to assess relevance, 39 studies were deemed suitable for inclusion in this review (see Fig. 1) (13–51). The details of these studies, along with a summary of their respective findings, are presented in Annex 1. Among the reviewed studies, 28 were journal articles, eight were reports published on specific organizations' websites, and three were conference articles. Notably, 27 out of the 39 studies were published within the last five years, suggesting growing scholarly attention towards the field of digital health competency.

Fig 1. Flow diagram for the narrative review search



The main findings can be broadly classified into the following subheadings.

3.1 Definition of a digital health competency framework

In the findings of the literature analysis, very few attempts had been made to provide a definition or full understanding of the term “digital health competency framework”. For instance, only a handful of studies ($n = 3$) attempted to define “digital health”, often citing the WHO interpretation; with even fewer addressing “eHealth” ($n = 1$). Nine studies ventured to clarify “competency” or “competence” in the context of digital health, generally encompassing a range of aspects such as knowledge, skills, attitudes, values and the capabilities necessary for proficient and effective participation in digital health-related endeavours. Notably, only one of the reviewed publications concretely articulated a “competency framework” as such, characterizing it as “...the essential set of competencies required to achieve competence in a specific area” (22).

3.2 Characteristics of digital health competency frameworks

The analysis identified 35 unique frameworks, with subsets recurrently employed across multiple studies. In 17 studies, researchers adapted pre-existing frameworks, while 22 studies introduced new frameworks designed for digital health competencies. The developers of these new frameworks frequently highlighted the lack of models specifically tailored to unique health worker demographics or particular digital health sectors. In contrast, those who adapted pre-existing frameworks refined their versions to align with the unique needs of specific health worker groups or digital health sectors.

Six studies proposed frameworks with an international scope, acknowledging the global significance and adaptability of digital health competencies. The TIGER (Technology Informatics Guiding Education Reform) International Recommendation Framework of Core Competencies was the most frequently utilized, cited by three studies. The TIGER Framework was developed with contributions from experts in 51 countries to establish core health informatics competencies for nurses (32,33) which were later adapted for other health professions (34). Similarly, the Health Information Technology Competencies (HITCOMP) Framework, a collaborative effort by experts from the European Union, United States of America and other nations, aimed to create a universal structure to enhance the competencies of health workers in digital health (29). Another international framework has been established, incorporating global inputs and perspectives to inform the best practices and competencies necessary for physiotherapists to deliver videoconferencing services on a global scale (23). While these internationally focused frameworks aimed for harmonization and cross-border knowledge exchange, they may not have fully addressed the specific nuances and needs of different regions and health care systems. Conversely, 20 frameworks focused on a localized approach, meeting the unique requirements of individual countries or regions, such as Australia, Canada, Germany, the Netherlands (Kingdom of the), New Zealand, Saudi Arabia, Switzerland, the United Kingdom of Great Britain and Northern Ireland, and the United States of America. For example, the DACH (Germany, Austria and Switzerland) digital competency framework was instrumental in classifying and ranking nursing competencies and enabling a detailed examination of the distinctions between country contexts and the specific competencies required (25,26).

The frameworks identified were noted to serve two principal functions: educational and practical. Those developed mainly for educational purposes aimed to ensure and clarify the competencies required by health workers, supporting competency-based medical education initiatives ($n = 15$). In the context of education, activities such as self-evaluation, self-directed learning, curriculum development and competency assessment were reported with the goal to inform curriculum and learning programmes. Other frameworks were noted to focus on competencies needed for practical applications, such as equipping job candidates, providing vocational guidance, and facilitating digital skills training ($n = 18$). A subset of frameworks ($n = 6$) identified the translational competencies necessary to bridge educational settings and clinical practice, reflecting the diversity of approaches where educational settings may prioritize knowledge acquisition over the skills and behaviours required for effective practice.

An examination of the digital health competency frameworks analysed for this report revealed that competencies (where mentioned or attributed) were typically categorized into multiple domains, dimensions or areas. These frameworks, however, varied significantly in their approach, including the number of dimensions (ranging from 3 to 33) and competencies (ranging from 5 to 1025), hierarchical structure (spanning 2 to 4 levels), and levels of descriptive detail (e.g. proficiency levels, roles, levels of practice, capability levels).

Additionally, an analysis of applied professions indicated a considerable number of digital health competency frameworks were nursing-centric, with 20 studies dedicated to the nursing profession, three of which also include midwifery. Some frameworks identified were designed for interprofessional use ($n = 10$), highlighting the collaborative nature of digital health and the need



for unified competency development among different health worker groups. The frameworks also differed in educational level; some are intended to instil digital health competencies at the pre-service level (n = 14), while others aimed to enhance these skills among in-service practitioners (n = 17). A select few frameworks did not differentiate between pre-service and in-service stages (n = 8).

The analysis of the included frameworks suggests that the 10 identified competency clusters or domains are intended for digital health or eHealth applications in general, while the remaining 29 frameworks address competencies specific to digital health specialties such as health informatics, telehealth and electronic health records (EHR). For a deeper exploration of the characteristics identified in this analysis of digital health competency frameworks, please refer to Annex 2.

3.3 Mapping competencies across frameworks

The mapping exercise for this report involved directly comparing competency domains, categories and areas present in 31 frameworks. Eight frameworks were excluded for various reasons (detailed in Annex 3). While most frameworks employed a domain-based structure, a few utilized the terms “categories” or “areas.” Notably, these terms used across frameworks seemed interchangeable. In essence, these frameworks defined domains/categories/areas as broad, distinct areas of core competencies that collectively form a comprehensive professional framework. Therefore, the terms were treated synonymously during the mapping process. Mapping these frameworks revealed 292 unique competency domains/categories/areas with some overlap. A process of synonym organization to code these was then employed, leading to the identification of 10 clusters as follows: research, informatics, data, digital professionalism/ safety and security, teaching and learning, technical proficiency, information systems and information and communication technologies (ICT), patient care, communication, and administration. Descriptions of each cluster are provided in Table 1, with full mapping details in Annex 4. A comparison of coverage frequency across competencies showed the most prevalent competencies to be patient care (n = 25), digital professionalism/safety and security (n = 22), data (n = 19) and technical proficiency (n = 19). All four competencies appeared in 10 of the 31 frameworks. Conversely, research (n = 11), informatics (n = 11), and teaching and learning (n = 9) were mapped the least. Due to the closely linked descriptions of the data and information systems/ICT clusters, it is worthwhile to note that they could be merged or interpreted similarly in practice. Although some frameworks were tailored to specific groups of health workers and others to interprofessional groups, common competency areas emerged across them. This underscores the interdisciplinary nature of digital health, as reflected in the identified overarching domain clusters.

Table 1. Description of competency clusters in digital health competency frameworks

Cluster	Examples of competency domains/categories/ areas (from retrieved frameworks)	Description
Research	Research/biomedicine (29); Research (37); Creation, innovation and research (13)	This cluster included competencies related to the adept use of digital tools and technologies to enhance investigative efforts. This includes leveraging research principles and various research platforms and tools critical to health interventions and technologies, aiding in evidence-informed development, conduct and evaluation of digital health innovations and procedures.
Informatics	Management in informatics (33); Informatics knowledge (16); Informatics skills (16); Specific informatics skills (51)	This cluster involved competencies related to the essential skills for managing, analysing and applying health information and data, grounded in IT principles and tools. This field requires knowledge in managing health information to aid clinical decisions and improve health care delivery, quality and efficiency.
Data	Data analysis and knowledge creation (18); Working with data and analytical models (23); Interpretation of analysis of eHealth data (17)	This cluster was centred on the skills necessary for handling health-related data, which encompasses the collection, analysis, interpretation and safeguarding of data through digital means. Competencies identified include data stewardship, data governance, analytical skills and the ability to translate insights from data into practical strategies that enhance patient outcomes and health care system operations.

Cluster	Examples of competency domains/categories/ areas (from retrieved frameworks)	Description
Digital professionalism/ safety and security	Digital identity, well-being, safety and security (13); Ethics and legal concepts (19); Professionalism (30); Legal and regulatory issues (38,39); Evidence-based and ethical practice (30)	This cluster covered the skills needed to maintain ethical standards and secure practices in the use of digital health technologies. This includes understanding digital ethics, ensuring privacy and implementing security protocols to protect sensitive health information and uphold the integrity of health care digital infrastructures.
Teaching and learning	Education (37); Practice-based learning and improvement (28); Teaching, training and education (48); Learning techniques (34,48)	Essential skills in this cluster included utilizing digital tools for teaching and professional development within the health care sector. It involves crafting and disseminating interactive e-learning materials and managing digital learning environments to keep health care professionals abreast of emerging digital health trends.
Technical proficiency	Computer literacy (43); Technology skills (23); eHealth, telematics, telehealth (25,26,32–34,48); Assistive technologies (25,26,32–34,48); Decision support systems (5,26,32)	This cluster pertained to the essential skills needed to navigate various digital health technologies. Proficiency in handling diverse hardware, software and applications is crucial to enhance health care delivery, patient management and overall efficiency in clinical settings.
Information systems and information and communication technologies (ICT)	Information systems literacy (45); Information systems management (45); Information and communication and technology literacy (45); Information technology and digital health systems (35); Information and communication technologies (42)	Proficiency in this cluster included the ability to access, integrate and maintain systems to facilitate the uninterrupted operations of digital health applications.
Patient care	Clinical practice and applications (18); Patient-centric care through digital channels (35); Information-supported care (21,44,50)	Competencies in this cluster pertained to the essential skills needed to integrate digital health technology in augmenting patient care. It involves utilizing digital resources and systems to bolster clinical judgements, patient involvement and the synchronization of care, enabling health care providers to deliver appropriate and quality care.
Communication	Relational skills and communication (20); Collaborative and interprofessional skills (20); Digital communication (36); Communication (15)	This cluster was linked to competencies necessary for effective communication and cooperation through digital means in health care environments. It covers competencies related to digital interactions in the exchange of information, coordination and joint efforts in health care delivery.
Administration	Leadership and management (14); System and technology implementation (18); Administration, policy (digital health infrastructures and investments) (37)	This cluster included competencies and the skills needed for the administration and management of digital health projects and initiatives. It involves expertise in project leadership, managing organizational processes and overseeing digital systems to confirm the effective adoption and refinement of digital health applications in health care institutions.

3.4 Methods employed for framework development in the included studies

Multiple methodological techniques were applied in the development of all the studies, with literature reviews or scoping exercises being the most common and appearing in 22 studies. Close behind were consultations with experts or stakeholders, presented in 20 studies, taking various forms such as focus groups, interviews, surveys, Delphi methods, workshops or feedback requests on a developed competency set. These groups typically had expertise in digital health, clinical practice, or specialized fields like telehealth or health informatics. Most of these experts were from academia, health organization leadership, professional associations and policy-making sectors. Notably, only two frameworks considered feedback from consumers or patients, indicating an opportunity for enhancing the framework's relevance and practicality for the intended users. Additional methods in developing these frameworks included adapting existing frameworks ($n = 8$), conducting curriculum mapping and content analysis ($n = 4$), and employing case studies or factor analysis for competency validation ($n = 3$). The development process often involved collaboration across various academic and non-academic entities ($n = 17$) or was spearheaded by single institutions ($n = 8$). Some were propelled by national or regional directives ($n = 10$) or through international agendas ($n = 4$).

Gaps associated in developing the frameworks ($n = 14$) reviewed were identified. One of the main issues identified during development of the frameworks reviewed was inadequate representation across diverse expert groups, varied gender dimensions, and minimal participation of professional organizations. Additionally, there was a critical absence of contributions from patients and consumers, who are the primary beneficiaries of these frameworks and whose insights and needs are vital to shaping effective digital health competencies.

Another notable gap found was linked to the framework application ($n = 3$). The lack of a unified or clear approach in these frameworks was evident due to a non-alignment and impossibility to compare them with other frameworks. Additionally, several key gaps have been identified at organizational and systems levels when introducing the frameworks for implementation ($n = 14$). A clear dearth of digital job role specifications and digital responsibility-specific appropriate health workforce training, resources and development programmes lowered the preparedness of health workers to contribute to digital transformation efforts. A lack of leadership (voice of reason and the promotion of digital readiness and agility) among those who can hasten digital transformation by championing change was also noted. Other gaps found include a lack of communication and motivational support from organizations to health workers.

Twelve studies recognized the potential for frameworks to quickly become obsolete due to the rapid evolution of digital health technologies, thus requiring regular updates. Efforts to address this gap were noted to involve not just revising the frameworks and the competencies they encompass, but also ensuring that their integration into education and professional practice considers their long-term viability. For instance, three studies reported attempts at long-term integration by ensuring implementation of frameworks in the development of educational programmes and training initiatives. Two of these studies concentrated on enhancing and creating teaching materials aligned with the desired digital health competencies. Another study reported staff self-use for evaluation of their digital competencies. Furthermore, one framework pursued official recognition from a health governing body to establish its acceptance as a vital benchmark for individuals assessing and validating their digital health proficiency.



4. Discussion

The first part of this section provides additional context to the findings reported in section 3. The second part relates more specifically to the applicability of the findings in practice; it introduces broader but related concepts and outlines the policy considerations that could support how stakeholders can best leverage digital health competency frameworks or standards to enhance learning or health workforce practice.

By synthesizing perspectives from the literature on digital health competency frameworks, this report draws particular attention to how these frameworks are defined. There is a broad understanding of digital health competence as an overarching concept that encompasses various clusters. The frameworks often include various hierarchical elements by breaking the concept into coherent categories. While explicit definitions of a digital health competency framework were absent within the studies reviewed, the term itself is situated within a broad semantic landscape filled with diverse and intersecting ideas. The construction of such frameworks often involves varied viewpoints and methodologies, with most adopting a hierarchical approach that begins by specifying broad domains, categories or areas, which are then subdivided into more focused subdomains. In this structured approach, competencies are then detailed typically alongside related abilities, skills, capacity statements, goals, practices and performance indicators. Additionally, some frameworks stratified competencies based on proficiency levels or roles, nuancing the requirements across different contexts or professional groups. Accordingly, competency frameworks tend to adopt heterogenic structures, and digital health competency frameworks are not exempt.

In essence, a digital health competency framework can be regarded as any schematic or narrative that seeks to clarify the understanding of digital health competencies by proposing an organizational schema for the concept. More specifically, it can be described as a synthesis of knowledge, skills, attitudes and other attributes necessary for the effective and responsible use of digital health technologies in tasks and problem-solving within increasingly digitalized health care learning and practice environments. However, such a seemingly overarching definition may obscure the inherent complexity and nuances of the various purposes or domains these frameworks cover.

The examination of the frameworks and competency domains of the reviewed sources affirms that TIGER, despite its primary nursing focus, provided a pivotal reference point for initiatives aimed at developing digital health competency frameworks. Several of these frameworks were either directly based on TIGER or considered it to varying extents. Nonetheless, all reviewed frameworks differed considerably in content and scope, with variations in competency dimensions and competencies influenced by factors such as individual circumstances of the health worker (e.g. roles, profession, education level and prior or foundational knowledge) and framework applications. Additionally, the digital health practice activities that frameworks underpin often require varying levels of competence to carry out. The use of DICOM (Digital Imaging and Communications in Medicine) standards for medical imaging systems by radio-diagnosticians, and of Epic and Cerner (nursing information systems) by nurses, are two examples illustrating this point (19,26,31,42,51). Hence, digital health competencies vary across and within occupational groups, and it is worthy, especially within interprofessional settings, to understand and leverage any such differences or similarities in designing how such competencies may be acquired and used by learners.

While key competencies vary based on previously mentioned factors, certain “building blocks” or competencies are consistent across most disciplines. The mapping and analysis of the domains across the 31 frameworks catering to core digital health tasks revealed the existence of 10 competency clusters that form the foundational basis of digital health practice. Patient care emerged as a consistent element throughout the examined frameworks, despite slight variations in terminology and concepts. This highlights the importance of preparing the health workforce with the knowledge of digital technology and the digital skills required for good patient care. Nonetheless, one challenge in synthesizing these competencies is the framing (e.g. descriptors, specification and levels of aggregation) by which competencies are articulated across the frameworks. Disparities exist in how the frameworks categorize these clusters of competencies, often referring to them interchangeably as domains versus categories versus areas. Such variations illustrate the diverse methods of documentation and the granularity of information provided by the different frameworks. Yet, summarizing and synthesizing these frameworks are seen to be of help in clarifying the competency requirements.

There is a clear need for widely recognized digital health competency frameworks or standards that are adaptable across a wide variety of health care settings. A more integrated, consensus-building approach among stakeholders has been fostered to address this challenge. While tailoring to local contexts remains important for adding nuance to action plans, having an overarching general framework is better appreciated at a higher level because it provides educators, policy-makers and employers with common pathways and a shared understanding of digital health competencies. Such frameworks could elucidate how these competencies apply to health workers across a variety of learning environments, clinical practice and employment settings.

The application of a set of broadly applicable digital health competencies in workforce development can further extend the opportunity to standardize learning and increase the likelihood that the digital skills of mobile health workers can be transferable across a wider variety of practice settings. However, it is recognized that global frameworks may not be directly applicable to practice at the country level if significant contextual differences are present locally. Some contextual considerations for adapting digital health competency standards can include economic viability, the existing maturity of integration of digital health technologies within the health and education systems, health governance, and the purpose for which digital health competency standards are being established and applied. For instance, in the present global economic climate, high-income countries are more likely to introduce advanced digital health technologies faster, whereas low-income countries may lack the funds, capacity or support to be as quick or agile (52). This gap in capacity between resource-rich and -low settings is commonly referred to as the phenomenon of the “digital divide”. Similarly, governments adopting a centralized approach may require digital health competency standards to be uniform across their health systems. On the other hand, a decentralized approach can allow regional structures to determine the relevant competencies and skills for their specific area and needs (53). As a result, decentralized systems may have a distinct set of prioritized digital health competency standards and accompanying digital health learning and practice tools. Decentralized systems are also likely to be impacted by interoperability issues and care must be taken to ensure alignment with other relevant systems and tools.

4.1 Fostering acquisition of digital health competencies

There is an increasing demand for health workers who are familiar with digital technologies and can use them safely and efficiently to offer or improve health services. Several countries have enacted legislative initiatives focused on building digital skills for their health workers. For example, Nordic countries like Denmark, Finland and Sweden have included legal requirements to integrate digital skills into the core curricula for educating and training all health professionals (54). The governments of Norway, Switzerland and the United Kingdom have similarly started initiatives, providing funding for interdisciplinary expert reviews to inform updates to health education curricula (54). eHealth Suisse plays a key role in advancing national initiatives on eHealth education through close collaborations among educational institutions, professional associations and health care organizations in Switzerland (55). In Canada, there have been concerted efforts across provinces and territories to engage health care institutions for the widespread adoption of digital health systems (54). Efforts like these across different countries demonstrate the growing global emphasis on establishing digital health competency standards to upskills the health care workforce. Moreover, in Australia, training providers, health care organizations and universities are working together to ensure future employees like health workers and project managers have the needed digital knowledge and skills to function optimally (56). The United Kingdom government also recently launched a 1-year postgraduate diploma in digital health leadership to upskill National Health Service managers and chief clinical information officers, focusing on areas like leadership and change management, health informatics and data analytics (57). WHO has also recently published a data management competency framework as an essential tool for health information management teams in Member States in the WHO Western Pacific Region to identify capacity gaps and plan and drive the competency development of their informatics workers across different levels of their health institutions and organizations (58).

While these types of initiatives represent positive steps forward, investment in training the current health workforce on major digital service delivery systems is still relatively low, despite governments investing in developing digital health services and infrastructure overall (54). The 2019 Global Digital Health Index, which assesses the preparedness and adoption of digital health in 22 countries across six WHO regions, noted that many countries are not ready to embrace a more digitalized system due to the lack of architecture, health information exchanges and data standards (59). According to the Index, 20 of 22 countries provide no training in digital health at the pre-service level, and 17 of 22 countries did not provide training to enable the acquisition of digital health competencies at the in-service level (59). Furthermore, important digital health competencies, like those enabling the delivery of telehealth, were deemed desirable only for knowledge (rather than also the ability to perform telehealth) for all pre-service students (24). Newer digital technologies such as generative AI have now become very popular with significant opportunities to transform clinical education and practice. It is therefore pertinent that policy-makers and



educators are equipped (60) and in turn, they are preparing their practitioners and learners to “be ready”, as appropriate for their health needs, resources and settings (61).

To determine the required level of digital competence of health workers, their demonstration of any acquired competencies must be assessed. Such assessment involves the digital practice activities or tasks that encompass the knowledge (e.g. understanding of information and facts), skills (e.g. abilities to carry out specific tasks using technologies or digital tools) and attitudes (e.g. beliefs toward something with some degree of favour or disfavour), and the competencies to integrate them (62). Several contextual factors must be considered for effective assessment at the individual level, including being able to demonstrate the answers to what digital technology to use, for whom or in what scenario or setting, and when to use such tools or technology. Due to the dynamic nature of digital health and technology in general, the following factors should also be considered when evaluating digital health competence: the educational level of the health worker (e.g. pre-service or in-service), the context (e.g. rural or urban) where the skill is being used, and the purpose (e.g. education or communication) for using the technology as well as its expected outcomes (e.g. patient engagement). Ready and equitable access to digital tools, and the gender of the user may also be factors to consider when evaluating digital health competence. During pre-service education, assessment emphasizes the testing of competencies required for future practice, whereas in-service assessments focus primarily on maintaining competence and keeping up with evolving practice standards, approaches or digital tools (62). As the purposes for pre-service and in-service education can differ in terms of the length of the training period, the timing of the assessment could also influence the reliability of the assessment outcomes regarding the digital health competencies being substantiated. At the individual level, health workers must be more proactive to keep abreast of emerging developments to maintain the required level of digital health competence.

While individual contributions are necessary, the success of strengthening processes that foster the acquisition of digital health competencies and evaluating the efficiency of these processes is contingent upon the availability of organizational support and a robust governance system (63). Organizational level actors are usually health or health training institutions such as hospitals and colleges that must comply with government rules and regulations in order to provide learning or health services to the public. On the other hand, system level actors are usually characterized by their authority as guiding entities that make guidelines and enforce rules and procedures for other organizations to follow. These can include health regulatory agencies and accreditation or licensing bodies for instance.

At the organizational and institutional levels, the identification of digital health competencies is easier when formulated based on a clear understanding of what health services or learning activities will involve or the use of digital tools. Institutions and organizations should also lead this process by ensuring that digital learning models, tools or platforms to educate and train health workers have the necessary guidance and resources to help deliver a proficient health workforce, with flexible digital health competencies (as feasible and where appropriate) to adapt to different work settings, changing demands and needs. Health institutions and other providers of digitalized learning programmes should also provide clear guidance to health and care workers on learning pathways and their expected outcomes. At the system level, institutions need to be proactive in providing a thriving environment for digital innovations and monitoring the same to ensure that practice and learning are aligned with existing needs and technological capacity. By staying ahead of the curve, policy- and decision-makers can have more time and better agility to identify recipients’ needs, and the equivalent digital resources required to deliver them.

Finally, the safe and effective use of digital technologies for learning and practice can be linked to how much investment is being put into training on digital health, as well as building a comprehensive and integrated digital health system that is underpinned by the design and other inputs of existing health workers themselves (64). Notwithstanding, the rapid pace of change in the adoption and use of digital health tools means that gaps in health workforce competence are likely to always be in play. This challenge can be countered by the proactivity of health workers and health training or service-providing organizations to develop targeted training for lagging competencies in order to adapt and thrive. As digital health competencies evolve in response to technological advances and health system demands, it is important that the concepts of change, adaption and sustainability are ingrained in the process of reviewing or applying digital health competency standards and subsequently, learning curricula for health workers. *Change* as a concept here refers to a shift in culture or practice that may be underpinned by the reversal, elimination or introduction of new approaches, digital tools or elements of digital health competency standards. *Adaptation* refers to a typically evidence- or information-guided process of contextualizing digital health competency standards or their application in education, practice or employment to meet specific needs or goals. *Sustainability* refers to the continuous and consistent application of fundamental principles of digital health competency standards and the maintenance or accumulation of digital resources to support health workforce education and practice. Policy options are



outlined below to also support policy-makers and other relevant stakeholders on how digital health competency standards can be contextualized in ways that can therefore maximize local use and impact (see Table 2).

Table 2. Development or adaption concepts and policy considerations

Contextual concept	Policy considerations	Description
Change	Explore the most effective strategies for monitoring advances in digital technologies for health practice and learning.	This involves updating digital health competency standards or frameworks to match current practice activities and other digital needs in terms of timeliness and cost-effectiveness to implement. End-users' safety and care improvement should also be monitored while setting up a clear process to retire obsolete standards, technologies, digital practice activities and learning curricula. Any decision to adapt a learning technology or adopt a new one can also be subjected to a health workforce education technology assessment (HETA) (65) review as a scientific approach to support effective learning technology selection and use.
	Delineate internal and external stakeholder roles and responsibilities and fine-tune management, monitoring and coordination mechanisms for the development of digital health competency standards or frameworks.	Stipulates the process of managing, monitoring and communicating expectations among stakeholders involved in the development, adaptation or review of digital health competency standards. This also involves the alignment of structures and mechanisms that may be required for effective coordination of newly introduced concepts or changes.
	Create or adopt a relevant assessment system for measuring the demonstration of digital health competencies.	Continuous and proper assessment of learners and practitioners allows for digital health competency standards to be matched or surpassed, thereby helping educational systems to foster fit-for-purpose and future-ready learning/training programmes.
Adaptation	Consult with interprofessional groups and experts, and relevant multistakeholder groups to define and distinguish digital health competencies that apply to different health worker groups.	Based on the contexts (needs and circumstances of the health workers and health systems involved), setting (global/national), profession and stage of learning (pre-service/in-service) relevant digital health competencies pertaining to specific groups of health workers should be identified to develop relevant educational programmes and competent health workers. To map and utilize the digital skills landscape at a wider scale, feedback from more extensive stakeholder consultation processes is invaluable to the development of digital health competency standards and subsequent learning curricula and programmes.
	Build intelligence gathering and data analysis into digital health competency standards use and evaluation.	Ensure that institutions and health service providers have adequate data points and monitoring and evaluation processes built into their digital health competency frameworks application. For example, the role of digital competency standards in enhancing digital equality in terms of access and use among men and women may become more visible if specific data are deliberated collected and analysed.
Sustainability	Ensure that critical principles of digital transformation are embedded in the development of digital health competency standards.	This includes ensuring that defined digital health competencies are learner-centred and gender-responsive and that a level playing field is promoted to gain digital competence through tailored applications according to the strengths of the particular group. For programming and digital public health goods, exploring the use of open-access features and contributions to global digital repositories and initiatives are useful characteristics to consider during development.
	Emphasize behavioural attitudes toward learning as a gold standard, rather than the mere prioritization of "new" technology adoptions in itself.	Technology innovation and diffusion are occurring at a rapid pace and are more likely to occur when end-user acceptance is widespread. Policy-makers and researchers should shape the outcomes by emphasizing learning attitudes and user satisfaction while optimizing the digital technology modality and fine-tuning digital health competency standards for health workers.



4.2 Limitations

The review is subject to certain limitations in its approach. The selection process for the studies, including the inclusion and exclusion criteria, has influenced the outcome of this analysis. The search strategy and terms were tailored to focus on literature related to digital health competency frameworks, potentially omitting other important contributions that discuss digital health competencies outside of these frameworks or provide a more in-depth analysis of the evolving dynamics, sustainability and adaptation. Furthermore, excluding studies not written in English could mean missing out on relevant studies that offer significant insights. It is important to recognize that digital health competencies can emerge and also evolve in definition and scope. As such, potential competencies or success factors not covered in this review may emerge in the future. This aspect is also relevant regarding the low representation of gender and equality dimensions in the included frameworks, which may be better elaborated as broader literature begins to produce more evidence. Lastly, this review did not assess the quality of the research included, which may impact how the results are interpreted.



5. Conclusion

To ensure that the health workforce is competent to use digital tools and technologies for learning or for health services provision, in addition to the policy options and success factors discussed in earlier sections, policy-makers need to ensure that they are addressing gaps in the definition and contextualization of digital health competency standards at the national and global levels. This report examines the landscape of digital health competency frameworks and theoretical and practical considerations to ensure that they contribute appropriately to strengthening health workers' digital health competencies across all settings. To further the development and effective implementation of digital health competency standards, policy-makers and relevant stakeholders should also consider other critical success factors such as the allocation of research funding to well-designed, larger scale studies and trials that examine the impacts of digital transformation across occupational groups and competence levels. These findings will help to improve understanding of digital health competency standards and how health workers may derive stronger positive attitudes and satisfaction as they are applied in learning and practice.



References

All references were accessed on 25 August 2025.

1. Catwell L, Sheikh A. Evaluating eHealth interventions: the need for continuous systemic evaluation. *PLoS Med*. 2009;6(8):e1000126 (<https://doi.org/10.1371/journal.pmed.1000126>).
2. WHO guideline. Recommendations on digital interventions for health system strengthening. Geneva: World Health Organization; 2019 (<https://www.who.int/publications/item/9789241550505>).
3. Global diffusion of eHealth: making universal health coverage achievable. Report of the third global survey on eHealth. Geneva: World Health Organization; 2016 (<https://iris.who.int/handle/10665/252529>). Licence: CC BY-NC-SA 3.0 IGO.
4. The Sustainable Development Goals report. United Nations; 2016 (<https://unstats.un.org/sdgs/report/2016/>).
5. Global strategy on digital health 2020-2025. Geneva: World Health Organization; 2021 (<https://iris.who.int/handle/10665/344249>). Licence: CC BY-NC-SA 3.0 IGO.
6. Global initiative on digital health. Geneva: World Health Organization; 2024 (<https://www.who.int/publications/m/item/global-initiative-on-digital-health>).
7. The Topol review: preparing the healthcare workforce to deliver the digital future. Health Education England: National Health Service; 2019 (<https://topol.hee.nhs.uk>).
8. Flattening the curve on COVID-19: how Korea responded to a pandemic using ICT. Seoul: Government of the Republic of Korea; 2020 (https://www.mofa.go.kr/eng/brd/m_22596/view.do?seq=4&srchFr=&srchTo=&srchWord=&srcHtp=&multi_itm_seq=0&itm_seq_1=0&itm_seq_2=0&company_cd=&company_nm=).
9. Mian A, Khan S. Medical education during pandemics: a UK perspective. *BMC Med*. 2020;18(1) (doi: [10.1186/s12916-020-01577-y](https://doi.org/10.1186/s12916-020-01577-y)).
10. Aungst TD, Patel R. Integrating digital health into the curriculum – considerations on the current landscape and future developments. *J Med Educ Curric Dev*. 2020;7:238212051990127 (doi: [10.1177/2382120519901275](https://doi.org/10.1177/2382120519901275)).
11. The rise of the data-driven physician. Stanford Medicine 2020 health trends report. Stanford Medicine; 2020 (<https://med.stanford.edu/content/dam/sm/school/documents/Health-Trends-Report/Stanford%20Medicine%20Health%20Trends%20Report%202020.pdf>).
12. Nazeha N, Pavagadhi D, Kyaw BM, Car J, Jimenez G, Tudor Car L. A digitally competent health workforce: scoping review of educational frameworks. *J Med Internet Res*. 2020;22(11):e22706 (doi: [10.2196/22706](https://doi.org/10.2196/22706)).
13. A health and care digital capabilities framework. National Health Service England; 2018 (<https://digital-transformation.hee.nhs.uk/binaries/content/assets/digital-transformation/nhs-digital-academy/digital-literacy-capability-framework-2018.pdf>).
14. Almalki M, Jamal A, Househ M, Alhefzi M. A multi-perspective approach to developing the Saudi Health informatics competency framework. *Int J Med Inform*. 2021;146:104362 (<https://doi.org/10.1016/j.ijmedinf.2020.104362>).
15. Applied public health informatics competency model. Decatur, GA: Public Health Informatics Institute; 2016 (<https://phii.org/wp-content/uploads/2021/06/Applied-Public-Health-Informatics-Competency-Model.pdf>).
16. Ayres EJ, Greer-Carney JL, Fatzinger Mcshane PE, Miller A, Turner P. Nutrition informatics competencies across all levels of practice: a national Delphi study. *J Acad Nutr Diet*. 2012;112(12):2042-53 (doi: [10.1016/j.jand.2012.09.025](https://doi.org/10.1016/j.jand.2012.09.025)).
17. Barakat A, Woolrych RD, Sixsmith A, Kearns WD, Kort HS. eHealth technology competencies for health professionals working in home care to support older adults to age in place: outcomes of a two-day collaborative workshop. *Med 2.0*. 2013;2(2):e10 (doi: [10.2196/med20.2711](https://doi.org/10.2196/med20.2711)).
18. Brunner M, McGregor D, Keep M, Janssen A, Spallek H, Quinn D et al. An eHealth capabilities framework for graduates and health professionals: mixed-methods study. *J Med Internet Res*. 2018;20(5):e10229 (doi: [10.2196/10229](https://doi.org/10.2196/10229)).
19. Collins S, Yen PY, Phillips A, Kennedy MK. Nursing informatics competency assessment for the nurse leader: the Delphi study. *J Nurs Adm*. 2017;47(4):212-8 (doi: [10.1097/NNA.0000000000000467](https://doi.org/10.1097/NNA.0000000000000467)).



20. Crawford A, Sunderji N, López J, Soklaridis S. Defining competencies for the practice of telepsychiatry through an assessment of resident learning needs. *BMC Med Edu*. 2016;16(1) (doi: [10.1186/s12909-016-0529-0](https://doi.org/10.1186/s12909-016-0529-0)).
21. Cummings E, Moran G, Woods L, Almond H, Procter P, Makeham M et al. Methodology for the development of the Australian national nursing and midwifery digital health capability Framework. *Stud Health Technol Inform*. 2021;284:135-142 (doi: [10.3233/SHTI210685](https://doi.org/10.3233/SHTI210685)).
22. Davies A, Mueller J, Hassey A, Moulton G. Development of a core competency framework for clinical informatics. *BMJ Health Care Inform*. 2021;28(1):e100356 (doi: [10.1136/bmjhci-2021-100356](https://doi.org/10.1136/bmjhci-2021-100356)).
23. Davies L, Hinman RS, Russell T, Lawford B, Bennell K. An international core capability framework for physiotherapists to deliver quality care via videoconferencing: a Delphi study. *J Physiother*. 2021;67(4):291-7 (doi: [10.1016/j.jphys.2021.09.001](https://doi.org/10.1016/j.jphys.2021.09.001)).
24. Dzioba C, Lamanna J, Perry CK, Toerber-Clark J, Boehning A, O'Rourke J. Telehealth competencies: leveled for continuous advanced practice nurse development. *Nurse Educ*. 2022;47(5):293-7 (doi: [10.1097/NNE.0000000000001196](https://doi.org/10.1097/NNE.0000000000001196)).
25. Egbert N, Thye J, Schulte G, Liebe JD, Hackl WO. An iterative methodology for developing national recommendations for nursing informatics curricula. *Stud Health Technol Inform*. 2016; 660-4 (doi: [10.3233/978-1-61499-678-1-660](https://doi.org/10.3233/978-1-61499-678-1-660)).
26. Egbert N, Thye J, Hackl WO, Müller-Staub M, Ammenwerth E, Hübner U. Competencies for nursing in a digital world. Methodology, results, and use of the DACH-recommendations for nursing informatics core competency areas in Austria, Germany, and Switzerland. *Inform Health Soc Care*. 2019;44(4):351-75 (doi: [10.1080/17538157.2018.1497635](https://doi.org/10.1080/17538157.2018.1497635)).
27. eHealth competencies for undergraduate medical education. Association of Faculties of Medicine of Canada/Canada Health Infoway; 2014.
28. Hart A, Romney D, Sarin R, Mechanic O, Hertelendy AJ, Larson D et al. Developing telemedicine curriculum competencies for graduate medical education: outcomes of a modified Delphi process. *Acad Med*. 2022;97(4):577-85 (doi: [10.1097/ACM.0000000000004463](https://doi.org/10.1097/ACM.0000000000004463)).
29. HITCOMP. Health Information Technology Competencies; 2020.
30. Hilty DM, Crawford A, Teshima J, Chan S, Sunderji N, Yellowlees P et al. A framework for telepsychiatric training and e-health: competency-based education, evaluation and implications. *Int Rev Psychiatry*. 2015;27(6):569-92 ([10.3109/09540261.2015.1091292](https://doi.org/10.3109/09540261.2015.1091292)).
31. Honey M, Collins E, Britnell S. Guidelines: informatics for nurses entering practice. Auckland: University of Auckland; 2018 (https://cdn.ymaws.com/www.hinz.org.nz/resource/resmgr/resources/Guidelines_Informatics_for_n.pdf).
32. Hübner U, Shaw T, Thye J, Egbert N, Marin H, Ball M. Towards an international framework for recommendations of core competencies in nursing and inter-professional informatics: the TIGER competency synthesis project. *Stud Health Technol Inform*. 2016;228:655-9 (doi: [10.3233/978-1-61499-678-1-655](https://doi.org/10.3233/978-1-61499-678-1-655)).
33. Hübner U, Shaw T, Thye J, Egbert N, Marin H, Chang P et al. Technology Informatics Guiding Education Reform – TIGER. *Methods Inf Med*. 2018;57(S 01):e30-e42 (doi: [10.3414/ME17-01-0155](https://doi.org/10.3414/ME17-01-0155)).
34. Hübner U, Thye J, Shaw T, Elias B, Egbert N, Saranto K et al. Towards the TIGER international framework for recommendations of core competencies in health informatics 2.0: extending the scope and the roles. *Stud Health Technol Inform*. 2019;264:1218-22 (doi: [10.3233/SHTI190420](https://doi.org/10.3233/SHTI190420)).
35. Jarva E, Oikarinen A, Andersson J, Tuomikoski A-M, Kääräinen M, Meriläinen M et al. Healthcare professionals' perceptions of digital health competence: a qualitative descriptive study. *Nurs Open*. 2022;9(2):1379-93 (doi: [10.1002/nop2.1184](https://doi.org/10.1002/nop2.1184)).
36. Jidkov L, Alexander M, Bark P, Williams JG, Kay J, Taylor P et al. Health informatics competencies in postgraduate medical education and training in the UK: a mixed methods study. *BMJ Open*. 2019;9(3):e025460 (doi: [10.1136/bmjopen-2018-025460](https://doi.org/10.1136/bmjopen-2018-025460)).
37. Lo B, Nagle LM, White P, Kleib M, Kennedy MA, Strudwick G. Digital and informatics competencies: requirements for nursing leaders in Canada. *Healthc Manage Forum*. 2021;34(6):320-5 (doi: [10.1177/08404704211015428](https://doi.org/10.1177/08404704211015428)).
38. Maheu MM, Wright SD, Neufeld JD, Drude K, Hilty DM, Baker D et al. Interprofessional telebehavioral health competencies framework: implications for telepsychology. *Prof Psychol Res Pr*. 2021;52(5):439-48 (doi: [10.1037/pro0000400](https://doi.org/10.1037/pro0000400)).
39. Maheu MM, Drude KP, Hertlein KM, Lipschutz R, Wall K, Hilty DM. An interprofessional framework for telebehavioral health competencies. *J Technol Behav Sci*. 2018;2:190-210 (doi: [10.1007/s41347-017-0038-y](https://doi.org/10.1007/s41347-017-0038-y)).
40. McBride S, Thomas L, Decker S. Competency assessment in simulation of electronic health records tool development. *Comput Inform Nurs*. 2020;38(5):232-9 (doi: [10.1097/CIN.0000000000000630](https://doi.org/10.1097/CIN.0000000000000630)).



41. McKinstry C, Iacono T, Kenny A, Hannon J, Knight K. Applying a digital literacy framework and mapping tool to an occupational therapy curriculum. *Aust Occup Ther J.* 2020;67(3):210-217 (doi: [10.1111/1440-1630.12644](https://doi.org/10.1111/1440-1630.12644)).
42. Nagle LM, Crosby K, Frisch N, Borycki E, Donelle L, Hannah K et al. Developing entry-to-practice nursing informatics competencies for registered nurses. *Stud Technol Health Inform.* 2014;201:356-63 (doi: [10.3233/978-1-61499-415-2-356](https://doi.org/10.3233/978-1-61499-415-2-356)).
43. National informatics standards for nurses and midwives. Australian Nursing and Midwifery Federation, Federal Office; 2015 (https://www.anmf.org.au/media/kgdb4ozx/national_informatics_standards_for_nurses_and_midwives.pdf).
44. All-Ireland nursing and midwifery digital health capability framework. Dublin and Belfast: Office of Nursing and Midwifery Services; 2022 (<https://healthservice.hse.ie/filelibrary/onmsd/all-ireland-nursing-midwifery-digital-health-capability-framework.pdf>).
45. O'Connor S, Larue E. Integrating informatics into undergraduate nursing education: a case study using a spiral learning approach. *Nurse Educ Pract.* 2021;50:102934 (doi: [10.1016/j.nepr.2020.102934](https://doi.org/10.1016/j.nepr.2020.102934)).
46. Recommendations on a common framework for mapping health professionals' eHealth competencies. JAseHN (Joint Action to Support the eHealth Network); 2018.
47. Rutledge CM, O'Rourke J, Mason AM, Chike-Harris K, Behnke L, Melhado L et al. Telehealth competencies for nursing education and practice: the four P's of telehealth. *Nurse Educ.* 2021;46(5):300-5 (doi: [10.1097/NNE.0000000000000988](https://doi.org/10.1097/NNE.0000000000000988)).
48. Thyé J, Shaw T, Hüsers J, Esdar M, Ball M, Babitsch B et al. What are inter-professional eHealth competencies? *Stud Technol Health Inform.* 2018;253:201-5 (doi: [10.3233/978-1-61499-896-9-201](https://doi.org/10.3233/978-1-61499-896-9-201)).
49. Van Houwelingen CTM, Moerman AH, Ettema RGA, Kort HSM, Ten Cate O. Competencies required for nursing telehealth activities: a Delphi-study. *Nurse Educ Today.* 2016;39:50-62 (doi: [10.1016/j.nedt.2015.12.025](https://doi.org/10.1016/j.nedt.2015.12.025)).
50. Woods L, Cummings E, Dobroff N, Nowlan S, Almond H, Procter P et al. Intended use of the national nursing and midwifery digital health capability framework. In: Merolli M, Bain C, Schaper LK (eds). *Healthier lives, digitally enabled. Studies in Health Technology and Informatics Vol. 276.* Amsterdam: IOS Press; 2021:106-11 (doi: [10.3233/SHTI210018](https://doi.org/10.3233/SHTI210018)).
51. Zare Shahi M, Mirzaei S, Nasiriani K. Nursing informatics competencies in critical care unit. *Health Informatics J.* 2022;28(1):146045822210838 (doi: [10.1177/1460458222108383](https://doi.org/10.1177/1460458222108383)).
52. Cory N, Stevens P. Building a global framework for digital health services in the era of COVID-19. Washington, DC: Information Technology & Innovation Foundation; 2020 (<https://itif.org/publications/2020/05/26/building-global-framework-digital-health-services-era-covid-19/>).
53. Sreeramareddy CT, Sathyanarayana T. Decentralised versus centralised governance of health services. *Cochrane Database Syst Rev.* 2019;9:CD010830 (doi: [10.1002/14651858.CD010830.pub2](https://doi.org/10.1002/14651858.CD010830.pub2)).
54. Socha-Dietrich K. Empowering the health workforce to make the most of the digital revolution. Paris: Organisation for Economic Co-operation and Development; 2021 (https://www.oecd.org/en/publications/empowering-the-health-workforce-to-make-the-most-of-the-digital-revolution_37ff0eaa-en.html).
55. Leitfaden für bildungsverantwortliche – Ehealth-themen für gesundheitsfachpersonen. Bern: eHealth Suisse; 2017.
56. Health in the 21st century: putting data to work for stronger health systems. Paris: Organisation for Economic Co-operation and Development; 2019 (https://www.oecd.org/en/publications/health-in-the-21st-century_e3b23f8e-en.html).
57. Sood H, McNeil K, Keogh B. Chief clinical information officers: clinical leadership for a digital age. *BMJ.* 2017;358:j3295 (doi: [10.1136/bmj.j3295](https://doi.org/10.1136/bmj.j3295)).
58. Data management competency framework. Manila: WHO Regional Office for the Western Pacific; 2023 (<https://iris.who.int/handle/10665/367502>). Licence: CC BY-NC-SA 3.0 IGO.
59. The state of digital health 2019. Global Digital Health Index; 2019 (<https://digitalhealthmonitor.org/stateofdigitalhealth19>).
60. Ethics and governance of artificial intelligence for health. Geneva: World Health organization; 2021 (<https://www.who.int/publications/i/item/9789240029200>).
61. Preiksaitis C, Rose C. Opportunities, challenges, and future directions of generative artificial intelligence in medical education: scoping review. *JMIR Med Educ.* 2023;9:e48785 (doi: [10.2196/48785](https://doi.org/10.2196/48785)).
62. Khurana MP, Raaschou-Pedersen DE, Kurtzhals J, Bardram JE, Ostrwoski SR, Bundgaard JS. Digital health competencies in medical school education: a scoping review and Delphi method study. *BMC Med Educ.* 2022;22(1):129 (doi: [10.1186/s12909-022-03163-7](https://doi.org/10.1186/s12909-022-03163-7)).



63. Brommeyer M, Liang Z. A systematic approach in developing management workforce readiness for digital health transformation in healthcare. *Int J Environ Res Public Health*. 2022;19(21):13843 (doi: [10.3390/ijerph192113843](https://doi.org/10.3390/ijerph192113843)).
64. Siribaddana P, Hewapathirana R, Sahay S, Jayatilleke A, Dissanayake VHW. 'Hybrid doctors' can fast track the evolution of a sustainable e-Health ecosystem in low resource contexts: the Sri Lankan experience. *Stud Health Technol Inform*. 2019;264:1356-1360 (doi: [10.3233/SHTI190448](https://doi.org/10.3233/SHTI190448)).
65. Health workforce education technology assessment: background research report. Geneva: World Health Organization; 2023 (<https://iris.who.int/handle/10665/373345>). Licence: CC BY-NC-SA 3.0 IGO.



Annex 1: Study characteristics

Table A1.1 Study characteristics, presented by author, title, document type, journal/organization and objectives

First author, year	Title	Document type	Journal/organization	Objectives
Almalki, 2021	A multi-perspective approach to developing the Saudi Health informatics competency framework	Journal article	International Journal of Medical Informatics	Develop a health informatics competency framework to guide the Saudi Commission for Health Specialties to introduce a health informatics certification programme that meets local health care needs and is aligned with the national digital health transformation strategy.
Association of Faculties of Medicine of Canada, 2014	eHealth competencies for undergraduate medical education	Report	Association of Faculties of Medicine of Canada/ Canada Health Infoway	Develop eHealth competencies for undergraduate medical education in Canada.
Australian Nursing and Midwifery Federation, 2015	National informatics standards for nurses and midwives	Report	Australian Nursing and Midwifery Federation	Develop standards to follow and provide a measure by which all registered nurses, midwives and enrolled nurses can assess their competence in information technology and information management systems.
Ayres, 2012	Nutrition informatics competencies across all levels of practice: a national Delphi study	Journal article	Journal of American Academy of Nutrition and Dietetics	Define informatics competencies for the field of nutrition and dietetics; determine the assignment of each competency to the appropriate level of practice.
Barakat, 2013	eHealth technology competencies for health professionals working in home care to support older adults to age in place: outcomes of a two-day collaborative workshop	Journal article	Medicine 2.0	Discuss the competencies required by health care professionals working in home care with eHealth technologies such as remote telecare and ambient assisted living, mobile health and fall detection systems.
Brunner, 2018	An eHealth capabilities framework for graduates and health professionals: mixed-methods study	Journal article	Journal of Medical Internet Research	Develop a framework that could be used to guide health curriculum design based on current evidence and stakeholder perceptions of eHealth capabilities expected of tertiary health graduates.



First author, year	Title	Document type	Journal/organization	Objectives
Collins, 2017	Nursing informatics competency assessment for the nurse leader: the Delphi study	Journal article	Journal of Nursing Administration	Identify nursing informatics competencies perceived as relevant and required by nurse leaders.
Crawford, 2016	Defining competencies for the practice of telepsychiatry through an assessment of resident learning needs	Journal article	BMC Medical Education	Explore residents' learning needs with respect to telepsychiatry from both residents' and faculty members' perspectives.
Cummings, 2021	Methodology for the development of the Australian national nursing and midwifery digital health capability framework	Journal article	Studies in Health Technology and Informatics	Develop a national nursing and midwifery digital health capability framework.
Davies A, 2021	Development of a core competency framework for clinical informatics	Journal article	BMJ Health and Care Informatics	Examine how participants feel the framework should be applied and to whom; identify competencies that were considered "core" and how they fit in with an evolving profession.
Davies L, 2021	An international core capability framework for physiotherapists to deliver quality care via videoconferencing: a Delphi study	Journal article	Journal of Physiotherapy	Identify the core capabilities that physiotherapists need in order to deliver quality care via videoconferencing.
Dzioba, 2022	Telehealth competencies: leveled for continuous advanced practice nurse development	Journal article	Nurse Educator	Develop essential Advanced Practice Registered Nurse student competencies to advance the training of nurse practitioners and support curricular development
Egbert, 2016	An iterative methodology for developing national recommendations for nursing informatics curricula	Journal article	Studies in Health Technology and Informatics	Develop an iterative triple methodology to yield validated and country-specific recommendations for informatics core competencies in nursing.
Egbert, 2019	Competencies for nursing in a digital world. Methodology, results, and use of the DACH recommendations for nursing informatics core competency areas in Austria, Germany, and Switzerland	Journal article	Informatics for Health and Social Care	Display how such recommendations can be developed, what competency areas are most relevant in the three countries, and how the recommendations can be implemented in practice.
eHAction Joint Action Supporting the eHealth Network, 2018	Recommendations on a common framework for mapping health professionals' eHealth competencies	Report	European Commission	Propose an eHealth competencies conceptual framework for eHealth professionals.
Hart, 2022	Developing telemedicine curriculum competencies for graduate medical education: outcomes of a modified Delphi process	Journal article	Academic Medicine	Develop a set of proposed competencies for a telemedicine curriculum in graduate medical education.

First author, year	Title	Document type	Journal/organization	Objectives
Hilty, 2015	A framework for telepsychiatric training and e-Health: competency-based education, evaluation and implications	Journal article	International Review of Psychiatry	Propose competencies for trainees and clinicians, with telepsychiatry situated within the broader landscape of e-mental health care.
HITCOMP, 2020	Health Information Technology Competencies	Website	EU*US eHealth Work Project	Develop a framework outlining competencies and the essential skills necessary for eHealth and health IT professionals.
Honey, 2018	Guidelines: informatics for nurses entering practice	Report	University of Auckland	Develop guidelines to inform undergraduate nursing education that identify the key knowledge, skills and behaviours toward nursing informatics for nurses as they enter practice as a Registered Nurse.
Hübner, 2016	Towards an international framework for recommendations of core competencies in nursing and inter-professional informatics: the TIGER competency synthesis project	Journal article	Studies in Health Technology and Informatics	Define a framework of globally accepted core competencies in informatics among various roles nurses can hold and match these findings within country-specific needs.
Hübner, 2018	Technology Informatics Guiding Education Reform – TIGER	Journal article	Methods of Information in Medicine	Define and validate a framework of globally accepted core competency areas in health informatics and to enrich this framework with exemplar information derived from local educational settings.
Hübner, 2019	Towards the TIGER international framework for recommendations of core competencies in health informatics 2.0: extending the scope and the roles	Journal article	Studies in Health Technology and Informatics	Extend the scope of the framework of recommendations of health informatics core competencies beyond nursing and to include further professional roles that contribute to the success of health information systems.
Jarva, 2022	Healthcare professionals' perceptions of digital health competence: a qualitative descriptive study	Journal article	Nursing Open	Provide insight into health care professionals' lived experiences of digital health competence with the objective of improving the knowledge of how digital health competence is perceived by health care professionals.

First author, year	Title	Document type	Journal/organization	Objectives
Office of the Nursing and Midwifery Services Director, 2022	All-Ireland nursing and midwifery digital health capability framework	Report	Office of Nursing and Midwifery Services Director	Develop a nursing and midwifery digital health capability framework.
Public Health Informatics Institute, 2016	Applied public health informatics competency model	Report	Public Health Informatics Institute	Develop the applied public health informatics competency model that frames the knowledge, skills and abilities of the general public health workforce across core functions of public health
Rutledge, 2021	Telehealth competencies for nursing education and practice: the four P's of telehealth	Journal article	Nurse Educator	Describe the development of telehealth competencies for education and practice.
Thye, 2018	What are inter-professional eHealth competencies?	Journal article	Studies in Health Technology and Informatics	Investigate which eHealth competencies lie at the intersection of the individual groups of health professionals.
van Houwelingen, 2016	Competencies required for nursing telehealth activities: a Delphi-study	Conference article	Nurse Education Today	Identify the nursing telehealth entrustable professional activities (NT-EPAs) and the required competencies that can contribute to the development of nursing telehealth education.
Woods, 2021	Intended use of the national nursing and midwifery digital health capability framework	Conference article	Australian Digital Health Agency	Describe how individual nurses and midwives, organizations and education providers could use the newly developed national nursing and midwifery digital health capability framework to build digital health capability.
Zareshahi, 2022	Nursing informatics competencies in critical care unit	Journal article	Health Informatics Journal	Determine the required dimensions of nursing informatics for nurses in intensive care units.



Annex 2: Overview of competency frameworks

Table A2.1 Competency frameworks, presented by author, framework name, geographical application, purpose, structure, profession, education level and application area

First author, year	Name of framework	Geographical application	Purpose	Competency structure	Profession	Education level	Application area
Almalki, 2021	Saudi Health Informatics Competency Framework	Country (Saudi Arabia)	Practice	Domains: 6 Subdomains: 22 Competencies: 92	Health informatics	In-service	Informatics
Association of Faculties of Medicine of Canada, 2014	eHealth Competencies for Undergraduate Medical Education ^a	Country (Canada)	Education	Roles: 5 Milestones: 2 Competencies: 25	Interprofessional	Pre-service	Digital health
Australian Nursing and Midwifery Federation, 2015	National Informatics Standards for Nurses and Midwives	Country (Australia)	Practice	Domains: 3 Standards: 10 Cues: 53	Nursing	In-service	Informatics
Ayres, 2012	Academy of Nutrition and Dietetics Nutrition Informatics Competencies Framework	Country (United States of America)	Practice	Categories: 3 Competencies: 216 Levels of practice: 6	Allied health	Pre-service In-service	Informatics
Barakat, 2013	eHealth Technology Competencies for Health Professionals Working in Home Care to Support Older Adults to Age in Place ^a	Country (Kingdom of the Netherlands)	Practice	Categories: 5 Competencies: 5 Requirements: 6 Skills: 14	Nursing	Pre-service In-service	Digital health
Brunner, 2018	eHealth Capabilities Framework for Graduates and Health Professionals	Not specified	Education Practice	Domains: 4 Performance cues: 40	Nursing	Pre-service	Digital health
Collins, 2017	Nursing Informatics Competency Assessment	Not specified	Practice	Categories: 15 Competencies: 74	Nursing	In-service	Informatics
Crawford, 2016	Competencies for the Practice of Telepsychiatry ^a	Not specified	Education	Domains: 9 Competencies: 15 Professional roles: 5	Psychiatry	In-service	Telehealth
Cummings, 2021	Australian National Nursing and Midwifery Digital Health Capability Framework ^a	Country (Australia)	Practice	Domains: 5 Competencies: 15 Capability levels: 3	Nursing and midwifery	In-service	Informatics



First author, year	Name of framework	Geographical application	Purpose	Competency structure	Profession	Education level	Application area
Honey, 2018	Guidelines: Informatics for Nurses Entering Practice ^a	Country (New Zealand)	Education	Principles: 4 Competencies: 18	Nursing	Pre-service	Informatics
Hübner, 2016	TIGER (Technology Informatics Guiding Education Reform) Framework ^a	International	Practice	Competency areas: 24 Nursing roles: 5	Nursing	In-service	Informatics
Hübner, 2018	TIGER (Technology Informatics Guiding Education Reform) Framework ^a	International	Education	Domains: 6 Competency areas: 24 Nursing roles: 5	Nursing	Pre-service	Informatics
Hübner, 2019	TIGER (Technology Informatics Guiding Education Reform) International Recommendation Framework of Core Competencies in Health Informatics 2.0 ^a	International	Education Practice	Competency areas: 33 Health care professional roles: 6	Interprofessional	Pre-service	Informatics
Jarva, 2022	Health Professional Digital Perceptions of Health Competence	Not specified	Practice	Main categories: 6 Categories: 29 Subcategories: 62	Interprofessional	In-service	Digital health
Jidkov, 2019	Health Informatics Competencies in Postgraduate Medical Education and Training in the UK	Country (United Kingdom)	Education	Domains: 6 Competencies: 16	Medicine	In-service	Informatics
Lo, 2021	Digital and Informatics Competencies for Nursing Leaders in Canada	Country (Canada)	Practice	Domains: 5 Competencies: 10	Nursing	In-service	Informatics
Maheu, 2018	An Interprofessional Framework for Telebehavioral Health Competencies	Not specified	Education Practice	Domains: 7 Telebehavioral objectives: 51 Telebehavioral practices: 149 Competency levels: 3	Interprofessional	Pre-service, In-service	Telehealth
Maheu, 2021	Coalition for Technology in Behavioral Science (CTiBs) Interprofessional Framework for Telebehavioral Health Competencies	Not specified	Education Practice	Domains: 7 Telebehavioral objectives: 51 Telebehavioral practices: 149 Competency levels: 3	Interprofessional	Pre-service, In-service	Telehealth
McBride, 2020	Competency Assessment in Simulation of EHRs (CASE) Tool	Not specified	Education Practice	Domains: 15 Competency levels: 5	Nursing	Pre-service In-service	Electronic health records



Annex 3: List of studies and corresponding competency domains/categories/areas

Table A3.1 List of studies, presented by author; competency domain/category/area and organization

First author, year	Domains/categories/areas of competency	Organized by
Almalki, 2021	Core principles; Information and communication technology (ICT); Health sciences; Health data analytics; Education and research; Leadership and management	Domains
Australian Nursing and Midwifery, 2015	Computer literacy; Information literacy; Information management	Domains
Brunner, 2018	Digital technologies, systems and policies; Clinical practice and applications; Data analysis and knowledge creation; Systems and technology implementation	Domains
Crawford, 2016	Assessment skills; Relational skills and communication; Collaborative and interprofessional skills; Administrative skills; Medico-legal skills; Community psychiatry and community-specific knowledge; Knowledge of health systems	Domains
Cummings, 2021	Digital professionalism; Technology; Quality data; Information-supported care; Leadership and advocacy	Domains
Davies A, 2021	Health and well-being in practice; Information technologies and systems; Working with data and analytical methods; Enabling human and organizational change; Decision-making, leading informatics teams and projects	Domains
Davies L, 2021	Compliance; Patient privacy and confidentiality; Patient safety; Technology skills; Telehealth delivery; Assessment and diagnosis; Care planning and management	Domains
HITCOMP, 2020	Administration; Direct patient care; Engineering/information systems/ICT; Informatics; Research/medicine	Domains
Hubner, 2018	Data, information and knowledge; Information exchange and information sharing; Ethics and legal issues; Systems life cycle management; Management in informatics; Biostatistics and medical technology	Domains
Jidkov, 2019	Information governance and security; System use and clinician safety; Digital communication; Information and knowledge management; Patient empowerment; Emerging technologies	Domains
Lo, 2021	Administration; Clinical care; Education; Policy; Research	Domains
Maheu, 2018	Clinical evaluation of care; Virtual environment and telepresence; Technology, legal and regulatory issues; Evidence-based and ethical practice; Mobile health technologies; Telepractice development	Domains
Maheu, 2021	Clinical evaluation and care; Virtual environment and telepresence; Technology, legal and regulatory issues; Evidence-based and ethical practice, Mobile health technologies; Telepractice development	Domains
Nagle, 2014	Information and knowledge management; Professional and accountability; Information and communication technologies	Domains

First author, year	Domains/categories/areas of competency	Organized by
National Health Service, 2018	Information, data and content; Teaching, learning and self-development; Communication, collaboration and participation; Digital identity, well-being, safety and security; Technical proficiency; Creation, innovation and research	Domains
O'Connor, 2021	Health service literacy; Information and communication technology literacy; Information management, information systems literacy; Information systems management; Patient/citizen digital health literacy	Domains
Office of the Nursing and Midwifery Services Director, 2022	Digital professionalism; Leadership and advocacy; Data and information quality; Information enabled-care; Technology	Domains
Ayres, 2012	Computer skills; Informatics knowledge; Informatics skills	Categories
Barakat, 2013	ICT attitudes and skills; Interpretation of analysis of eHealth data; Support and guidance, communication skills, privacy and confidentiality	Categories
Collins, 2017	Management concepts; Requirements and system selection; Ethics/legal concepts; Information systems concepts; Advanced software applications; Executive leadership; Financial implementation/management; Patient-related applications; Data issues; Technical knowledge; Collaboration; Electronic communications; HIT selection; Standardization	Categories
Hart, 2022	Professionalism; Systems-based practice; Patient care; Practice-based learning and improvement; Interpersonal and communication skills; Medical knowledge	Categories
Jarva, 2022	Patient-centric care through digital channels; Information technology and digital health systems; Interacting with the patient through digital means; Evaluate what digital health is; Combine digital means and traditional methods; Evaluate their own digital health competencies	Categories
Public Health Informatics Institute, 2016	Principles and strategy; Standards and interoperability; Project management; Information systems; Communication; Evaluation; Analysis visualization and reporting; Policy	Categories
Woods, 2021	Digital professionalism; Leadership and advocacy; Data and information quality; Information enabled care, technology	Categories
Zareshahi, 2022	Basic computer skills; Skills to use information management software; Specific informatics skills	Categories
Egbert, 2016	Principles of nursing informatics; Applied computer science; Project management; Data protection and security; Nursing documentation; Information and communication systems for nursing; eHealth, telematics, telehealth; Information management in research; Information management in education and continuing education; Decision support systems; Image and biosignal processing; Quality assurance and management; Biostatistics/statistics; Resource planning and logistics; Assistive technologies; Ethics and IT; Principles of management; Strategic management and leadership; Change and stakeholder management; IT risk management; Financial management in nursing informatics; Human resource management in nursing informatics; Process management; Information and knowledge management in patient care	Areas
Egbert, 2019	Principles of nursing informatics; Applied computer science; Project management; Data protection and security; Nursing documentation; Information and communication systems for nursing; eHealth, telematics, telehealth; Information management in research; Information management in education and continuing education; Decision support systems; Image and biosignal processing; Quality assurance and management; Biostatistics/statistics; Resource planning and logistics; Assistive technologies; Ethics and IT; Principles of management; Strategic management and leadership; Change and stakeholder management; IT risk management; Financial management; Human resource management; Process management; Information and knowledge management in patient care	Areas
Hilty, 2015	Patient care; System-based learning; Practice-based learning; Professionalism; Communication; Knowledge; Technology	Areas



First author, year	Domains/categories/areas of competency	Organized by
Hubner, 2016	Principles of nursing informatics; Applied computer science; Project management, data protection and security; Nursing documentation; Information and communication systems for nursing; eHealth, telematic, telehealth; Information management in research; Information management in education and continuing education; Decision support systems; Image and biosignal processing; Quality assurance and management; Biostatistics/statistics; Resource planning and logistics; Assistive technologies; Ethics and IT; Principles of management; Strategic management and leadership; Change and stakeholder management; IT risk management; Financial management in nursing informatics; Human resource management in nursing informatics; Process management; Information and knowledge management in patient care	Areas
Hubner, 2019	Applied computer science; Assistive technology; Change/stakeholder management; Clinical decision support by IT; Communication; Consumer health informatics; Data analytics; Data protection and security; Documentation; e/mHealth, telematics, telehealth; Ethics in health IT; Financial management; Care processes and IT integration; ICT systems (applications); ICT systems (architectures); Information management research; Information and knowledge management in patient care; Interoperability and integration; IT risk management; Leadership; Learning techniques; Legal issues in health IT; Medical technology; Principles of health informatics; Principles of management; Process management; Project management; Public health informatics; Quality and safety management; Resource planning and management; Strategic management; System lifecycle management; Teaching, training, education	Areas
Thye, 2018	Process management; Clinical decision support by IT; Documentation; Information and communication technology/systems (applications); Information and knowledge management in patient care; eHealth, mHealth, telematics and telehealth; Health care processes and IT integration; Consumer health informatics; Public health informatics; Quality and safety management; Assistive technology for ageing people; Principles of health informatics; Interoperability and integration; Applied computer science; Data protection and security; Ethics in health IT; System lifecycle management; Medical technology; IT risk management; Information and communication technology/systems (architecture); Information management in research; Teaching, training and education in health care; Strategic management; Principles of management; Change and stakeholder management; Legal issues in health IT; Financial management; Resource planning and management; Project management; Data analytics; Communication; Leadership; Learning technique	Areas
Excluded from mapping		Justification
Dzioba, 2022	Planning; Preparing; Providing; Performance evaluation	This framework's domain was categorized by processes.
eHealth Joint Action Supporting the eHealth Network, 2018	Manage; Plan; Build; Use; Run; Enable	This framework's domain was categorized by processes.
McBride, 2020	Documentation is honest and demonstrates professional integrity; Documentation is devoid of inappropriate abbreviations, spelling errors, poor grammar and emoticons; Documentation is consistent with the assessed clinical condition; Documentation is complete with no gaps and no omissions; Appropriate use of structured or unstructured data within the EHR for collecting valid and reliable data that can be integrated into other assessment parameters to support sound clinical reasoning; Documentation reflects sound clinical reasoning based on clinical findings and evidence-based practice appropriate to patient condition and clinical environment; Documentation aligns with professional scope of practice; Documentation reflects timely reassessment of the patient and other pertinent diagnostic data following interventions; Documentation reflects efficiency demonstrated by proficient use of EHR hardware and software with minimal clicks or keystrokes to accomplish the required documentation within the clinical workflow; Documentation and actions reflect appropriate response to clinical decision support and triggers/alerts from the EHR	This framework's domain was categorized by processes.



Annex 4: Framework mapping of corresponding domains/categories/areas

Table A4.1 Framework mapping, presented by name, source and area of focus

Source	Research	Informatics	Data	Digital professionalism/safety and security	Teaching and learning	Technical proficiency	Information systems and information communication technologies (ICT)	Patient care	Communication	Administration
Almalki, 2021 (D)	(Education and) research	Core principles (of informatics)	Core principles (of data) Health data analytics		Education (and research)		Information and communication technology	Health sciences		Leadership and management
Australian Nursing and Midwifery, 2015 (D)			Information literacy	Information management (safe/legal/ethical management of health information)		Computer literacy				
Brunner, 2018 (D)			Data analysis and knowledge creation			Digital technologies, systems and policies		Clinical practice and applications		System and technology implementation
Crawford, 2016 (D)				Medico-legal skills		Technical skills		Knowledge of health system Assessment skills Community psychiatry and community-specific knowledge	Relational skills and communication Collaborative and interprofessional skills	Administrative skills
Cummings 2021 ; Office of the Nursing and Midwifery Services Director, 2022; Woods, 2021 (D)			Quality data	Digital professionalism		Technology		Information-supported care		Leadership and advocacy

Source	Research	Informatics	Data	Digital professionalism/ safety and security	Teaching and learning	Technical proficiency	Information systems and communication technologies (ICT)	Patient care	Communication	Administration
Davies A, 2021 (D)	Health and well-being in practice (scientific and research skills)	Health and well-being in practice (informatics strategies/informatics in health) Leading informatics teams and projects	Working with data and analytical models				Information technologies and systems	Health and well-being in practice (clinical concepts/care delivery) Decision-making		Health and well-being in practice (health administration and services) Enabling human and organizational change
Davies L, 2021 (D)				Compliance Patient privacy and confidentiality Patient safety		Technology skills		Telehealth delivery Assessment and diagnosis Care planning and management		
HITCOMP, 2020 (D)	Research/ Biomedicine	Informatics					Engineering/ information systems/ ICT	Direct patient care		Administration
Hübner, 2018 (D)		Management in informatics	Data, information and knowledge Biostatistics and medical technology	Ethics and legal issues			Systems life cycle management		Information exchange and information sharing	
Jidkov, 2019 (D)			Information and knowledge management (use of data)	System use and clinician safety Information governance and security		Information and knowledge management (understanding technology) Emerging technologies		Patient empowerment	Digital communication	
Lo, 2021 (D)	Research				Education			Clinical care		Administration policy (digital health infrastructure and investments)



Source	Research	Informatics	Data	Digital professionalism/ safety and security	Teaching and learning	Technical proficiency	Information systems and communication technologies (ICT)	Patient care	Communication	Administration
Maheu, 2018; Maheu, 2021 (D)				Legal and regulatory issues Evidence-based and ethical practice		Mobile health technologies Telepractice development		Virtual environment and telepresence		
Nagle, 2014 (D)				Professional and accountability			Information and communication technologies	Information and knowledge management		
National Health Service, 2018 (D)	Creation, innovation and research		Information, data and content	Digital identity, well-being, safety and security	Teaching, learning and self-development				Communication, collaboration and participation	
O'Connor, 2021 (D)			Information management				Information systems literacy Information systems management Information and communication technology literacy	Health services literacy Patient/citizen digital health literacy		
Ayres, 2012 (C)		Informatics knowledge Informatics skills				Computer skills				
Barakat, 2013 (C)			Interpretation of analysis of eHealth data	Privacy and confidentiality			ICT attitudes and skills	Support and guidance	Communication skills	
Collins, 2017 (C)			Data issues	Ethics/legal concepts		Technical knowledge Advanced software applications	Requirements and system selection Information systems concepts	Patient-related applications Health information technologies selection	Collaboration Electronic communications	Management concepts Executive leadership Financial Implementation/ management standardization



Source	Research	Informatics	Data	Digital professionalism/ safety and security	Teaching and learning	Technical proficiency	Information systems and communication technologies (ICT)	Patient care	Communication	Administration
Hart, 2022 (C)				Professionalism	Practice-based learning and improvement			Systems-based practice Patient care Medical knowledge	Interpersonal communication and skills	
Jarva, 2022 (C)						Combine digital means and traditional methods Evaluate what digital health is Evaluate their own digital health competence	Information technology and digital health systems	Patient-centric care through digital channels	Interacting with the patient through digital means	
Public Health Informatics Institute, 2016 (C)			Analysis visualization and reporting	Policy (laws and regulations)			Standard and interoperability Information systems		Communication	Project management Evaluation Principles and strategy
Zarehshahi, 2022 (C)		Specific informatics skills				Basic computer skills Skills to use information management software				
Egbert, 2016; Egbert, 2019; Hubner, 2016 (A)	Information management in research	Principles of nursing informatics	Image and biosignal processing Biostatistics/statistics	Data protection and security Ethics and IT IT risk management	Information management in teaching, training and education	eHealth, telematics, telehealth Assistive technologies Decision support systems	Applied computer science Information and communication systems for nursing	Information and knowledge management in patient care	Principles of management Financial management Process management Project management Nursing documentation Strategic management and leadership Resource planning and logistics Change and stakeholder management Quality assurance and management Human resource management	



Source	Research	Informatics	Data	Digital professionalism/ safety and security	Teaching and learning	Technical proficiency	Information systems and information communication technologies (ICT)	Patient care	Communication	Administration
Hubner, 2019; Thye 2018 (A)a	Information management research	Principles of health informatics Consumer Health informatics Public health informatics	Data analytics	Data protection and security IT risk management Ethics in health IT Legal issues in health IT	Teaching, training, education Learning techniques	e/mHealth, telematics, telehealth Medical technology Assistive technology Clinical decision support by IT	Applied computer science ICT systems (architectures) ICT systems (applications) Interoperability and integration	Information and knowledge management in patient care Health care processes and IT integration	Communication	Principles of management Financial management Process management Project management Documentation Strategic management Resource planning and management Leadership Change/stakeholder management Quality and safety management System lifecycle management
Hilty, 2015 (A)	Knowledge (evidence-based/clinical guidelines)			Professionalism	Systems-based learning Practice-based learning			Patient care	Communication (Communication using technology)	

Notes: D Domains; C Categories; A Areas.

